

BME1322: Artificial Intelligence (BME)

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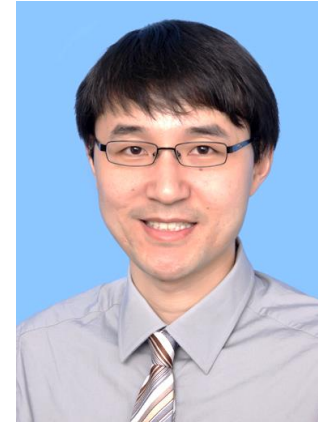
Introduction to Myself

- Research Interest

- Integrate AI with medical image computing;
- Smart diagnosis and prognosis;
- Individualized therapy planning and tracking;
- Translational medical studies.

- Experience

- 2021- BME @ShanghaiTech
- 2013-2021 BME @SJTU
- 2009-2013 PhD, CS @UNC Chapel Hill
- 2002-2006 BS/MS, EE @SJTU



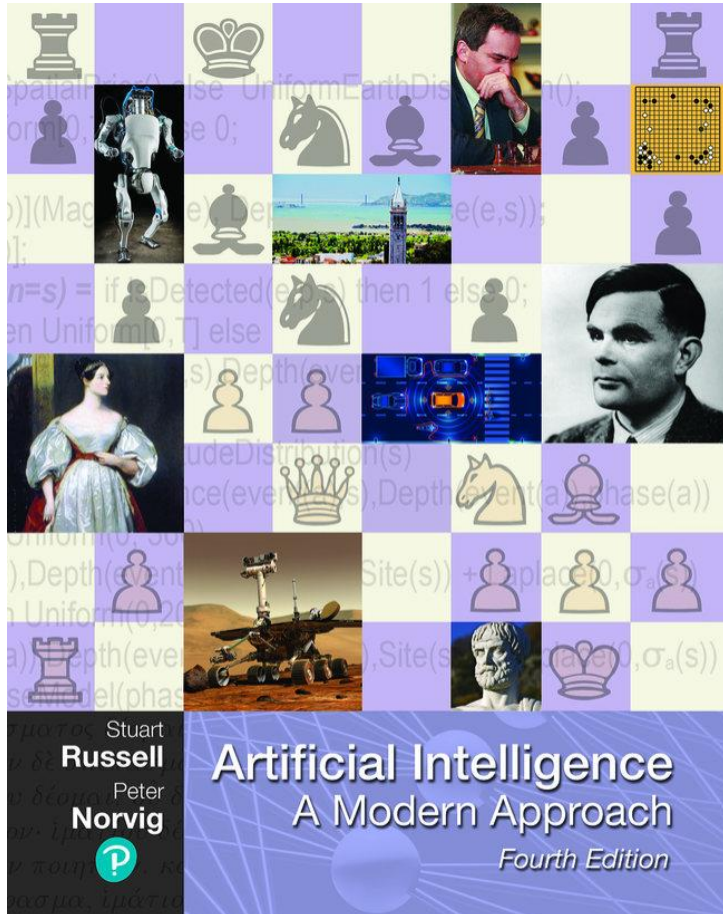
<https://qianwang.space>

Aims of this Course

- BME1322 – an AI course for multidisciplinary students
 - This course offers a comprehensive introduction to AI with a focus on its integration across multiple disciplines, particularly within biomedical engineering.
- Students will:
 - Gain a deep understanding of AI fundamentals, including machine learning and neural networks.
 - Develop hands-on skills in designing, training, and deploying AI models and systems.
 - Apply AI methodologies to solve real-world challenges in biomedical contexts, such as medical imaging, computer-assisted diagnosis, and personalized treatment planning.

This course reflects the growing trend of “AI + X” across diverse fields.

Textbook

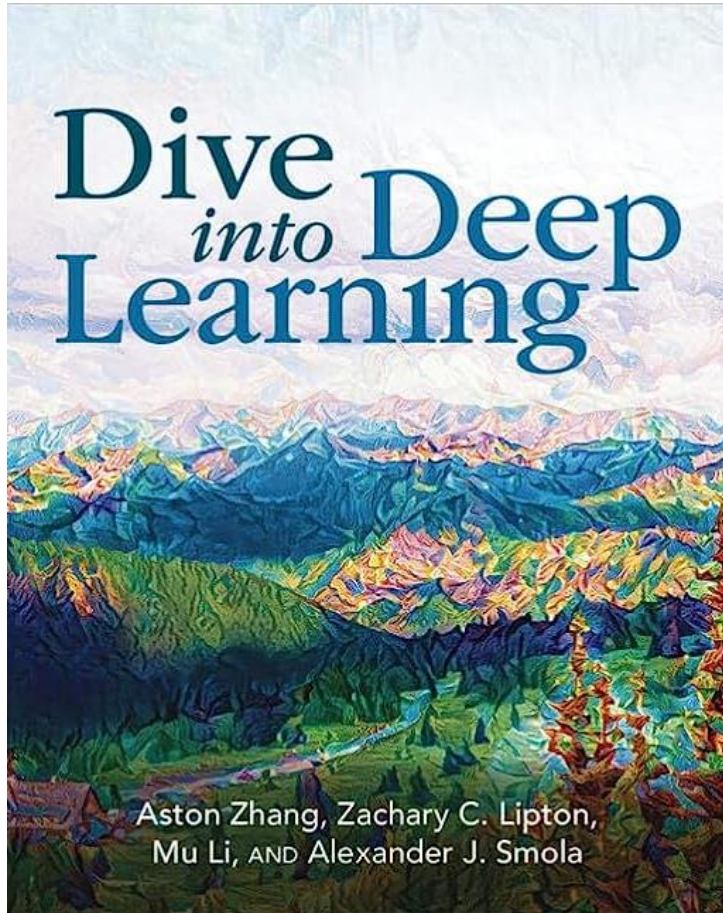


Artificial Intelligence: A Modern Approach, 4th US ed.
by *Stuart Russell* and *Peter Norvig*

<https://aima.cs.berkeley.edu/>

Materials and handouts will be available through E-learning system.

Textbook



Dive into Deep Learning
by *Aston Zhang* et al.

<https://d2l.ai/>

Materials and handouts will be available through E-learning system.

Grading

- Quiz: 10%
- Assignment & Project: 70%
- Final Exam: 20%
- Contact me for office hour when you need
 - qianwang AT shanghaitech.edu.cn
 - 410, BME Bldg.

Introduction to AI

Start of AI

- The first time “artificial intelligence” adopted was in Dartmouth Conference of 1956.



John McCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff



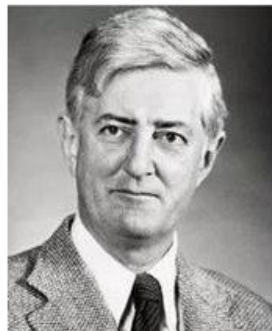
Alan Newell



Herbert Simon



Arthur Samuel



Oliver Selfridge



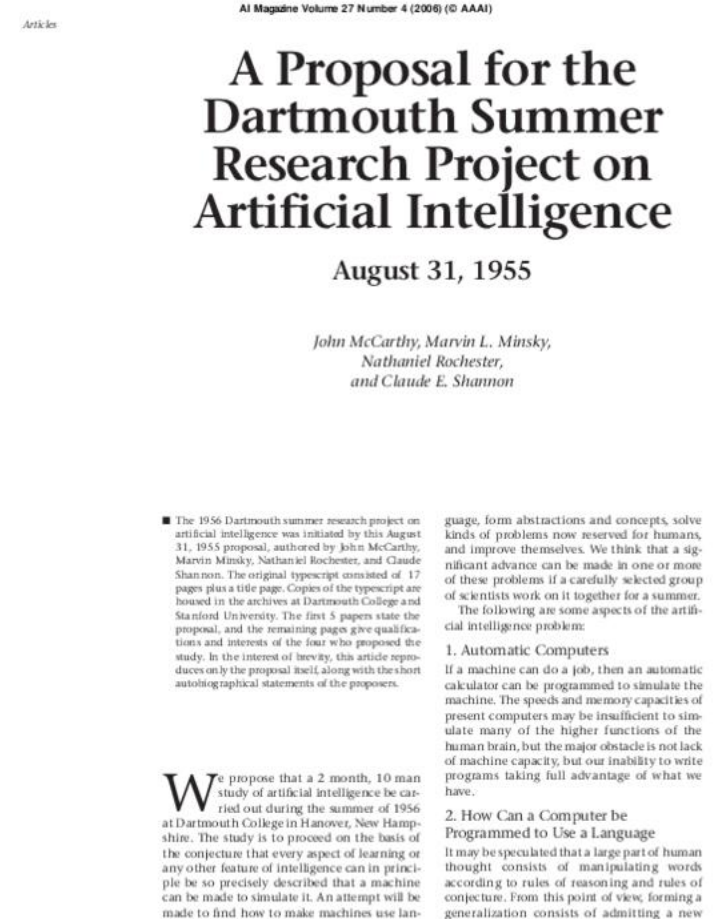
Nathaniel Rochester



Trenchard More

Initial Questions of AI

- Simulating human intelligence with machines
 - Automatic computers
 - Use a language to program a computer
 - Neuron networks
 - Theory of the size of calculation
 - Self-improvement
 - Abstractions
 - Randomness and creativity



Harder than Thought



Herbert Simon

Turing Award 1975

Nobel Prize 1978

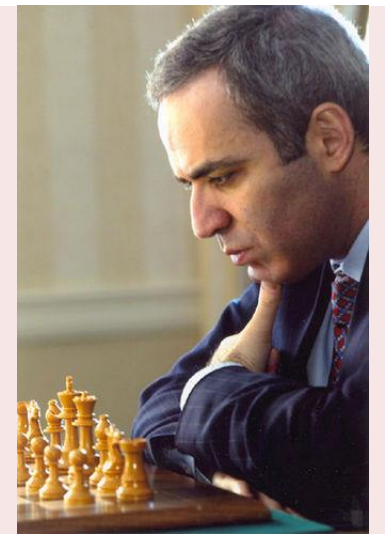
“It is not my aim to surprise or shock you but ... there are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until in a visible future – the range of problems they can handle will be coextensive with the range to which human mind has been applied. More precisely: within 10 years a computer would be chess champion, and an important new mathematical theorem would be proved by a computer.”

Prediction came true, but 40 years later:

In 1997, IBM's Deep Blue became the first computer to beat a chess champion when it defeated Russian grandmaster Garry Kasparov.



VS



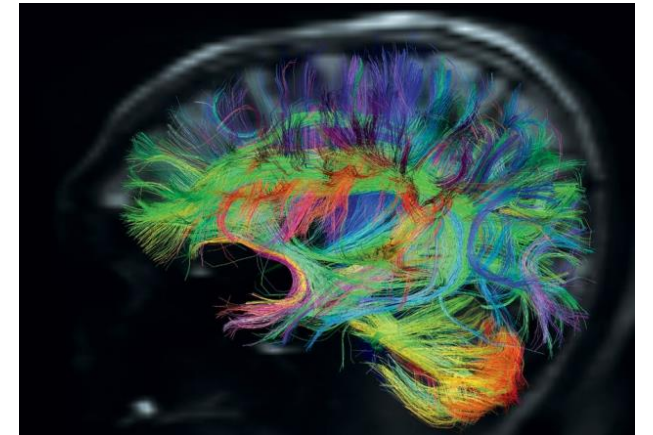
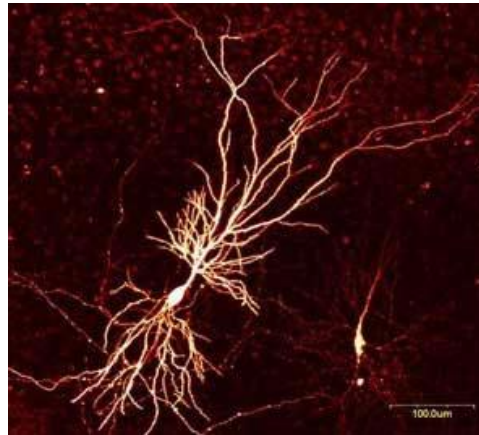
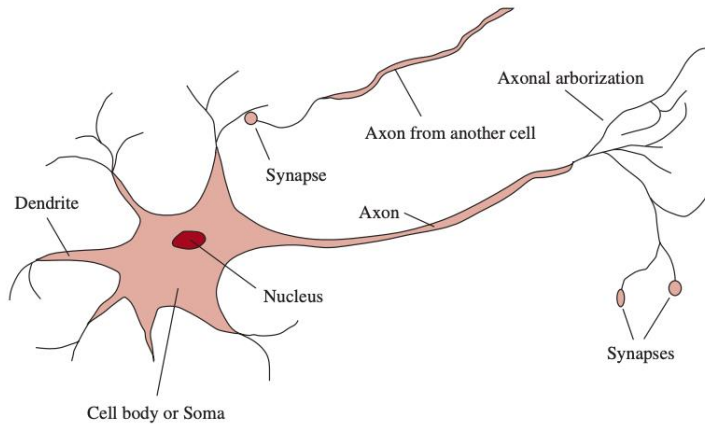
What is AI?

AI is the science of making machines that are:

1. Thinking humanly	2. Acting humanly
3. Thinking rationally	4. Acting rationally

Thinking Humanly

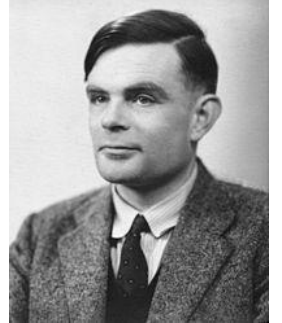
- Cognitive modeling: Make machines "think like humans"
 - Mimics how humans perceive, reason, and make decisions;
 - Based on models from psychology and neuroscience;
 - Focuses on replicating human learning and problem-solving behavior.



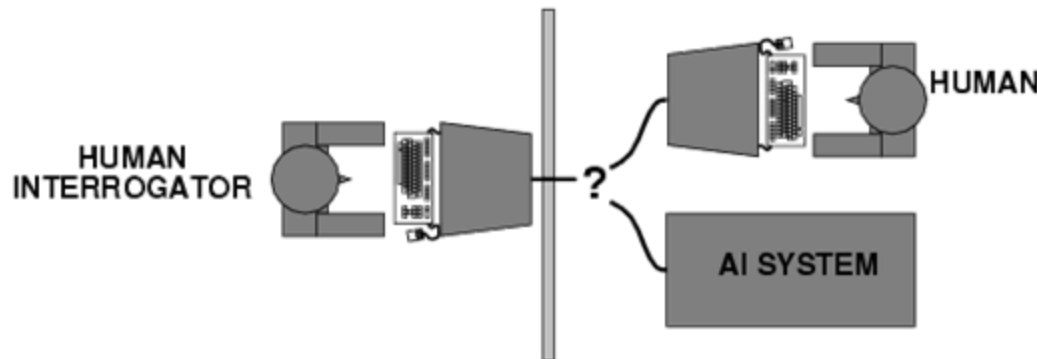
We can simulate parts of human thinking now, but building a full human brain remains far beyond our reach.

Acting Humanly

- Turing Test: Design AI systems that behave in ways indistinguishable from humans.
 - Interact with the world in a human-like manner;
 - Capabilities include speech, movement, and social interaction;
 - Focus is on external behavior, not internal reasoning.



Alan Turing



- An AI engages in conversation with a human evaluator and attempts to appear human through its responses.
- If the evaluator cannot reliably tell whether they're speaking to a human or a machine, the AI is considered intelligent.

Passing the Turing Test means behaving so naturally that humans can't tell it's a machine.

Acting Humanly

- What capabilities would a computer need to have to pass the Turing test?

*The sack of potatoes had been placed **below** the bag of flour, so it had to be moved first. What had to be moved first?*

- *The sack of potatoes*
- *The bag of flour*

*The sack of potatoes had been placed **above** the bag of flour, so it had to be moved first. What had to be moved first?*

- *The sack of potatoes*
- *The bag of flour*

Acting Humanly



The sack of potatoes had been placed below the bag of flour, so it had to be moved first. What had to be moved first?

A. The sack of potatoes

B. The bag of flour

Make a selection from A or B.



I would select **A. The sack of potatoes.**

The sentence implies that the sack of potatoes was placed below the bag of flour and needed to be moved first in order to access the bag of flour.

Answer from ChatGPT-4o (September 2024)

Acting Humanly

The sack of potatoes had been placed below the bag of flour, so it had to be moved first. What had to be moved first?

- A. The sack of potatoes
- B. The bag of flour

Make a selection from A or B

B. The bag of flour

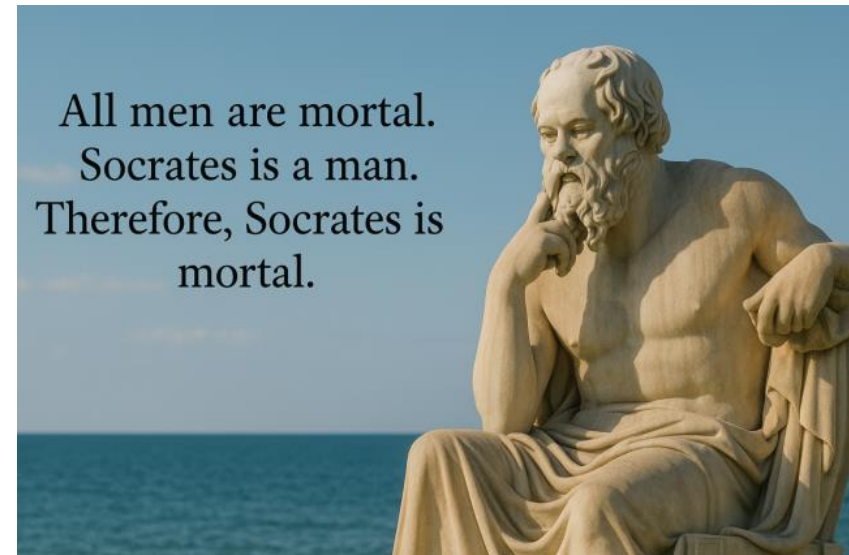
Because the sack of potatoes is *below* the bag of flour, the flour is on top. So to access or move the potatoes, the **bag of flour must be moved first**. This tests spatial reasoning and pronoun resolution—skills that are surprisingly tricky for AI systems! Want to try a harder one?

Answer from Copilot in Microsoft Edge
(based on some GPT-4 variant, September 2025)

The comparison shows how rapidly GPT-style models have improved in just one year.

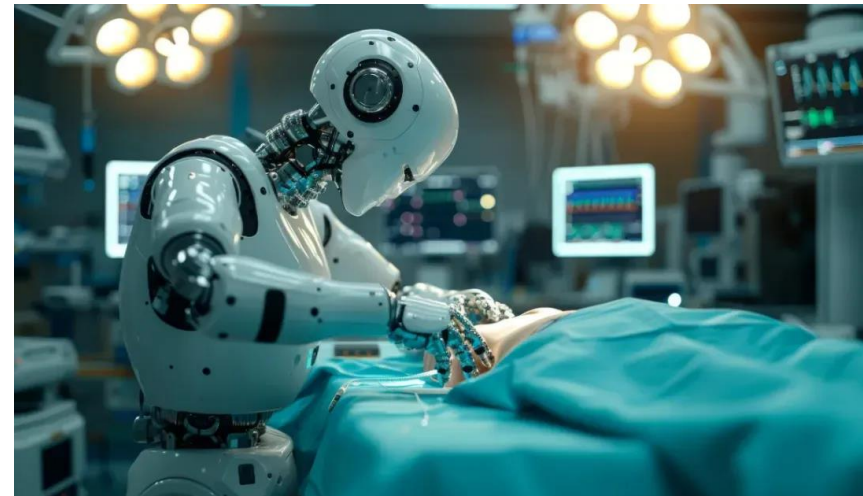
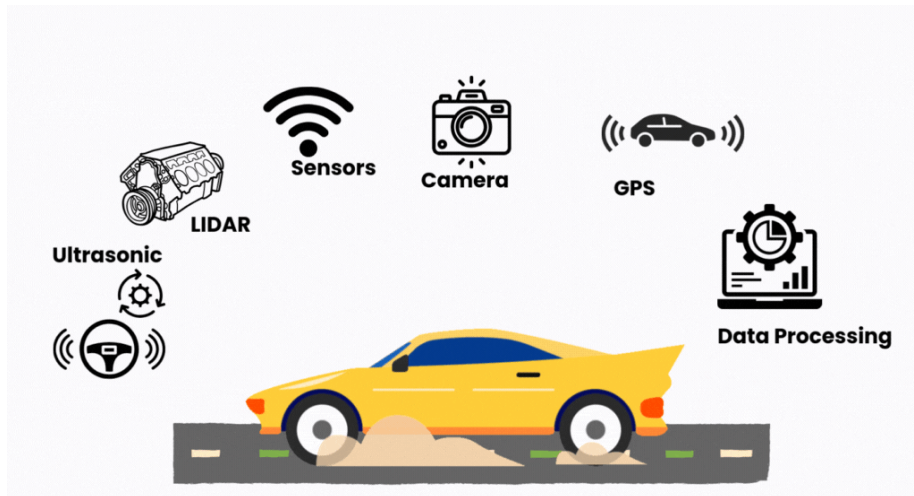
Thinking Rationally

- “Laws of Thought”: Build AI systems that make decisions by following formal rules of logic and mathematics.
 - Based on structured reasoning methods;
 - Aims for objectively correct conclusions, free from human bias or emotion;
 - Focuses on what should be done, given a set of premises.
- Classical logic example:
 - “All men are mortal”; Socrates is a man;
Premise 1 *Premise 2*
Therefore, Socrates is mortal.”
Conclusion



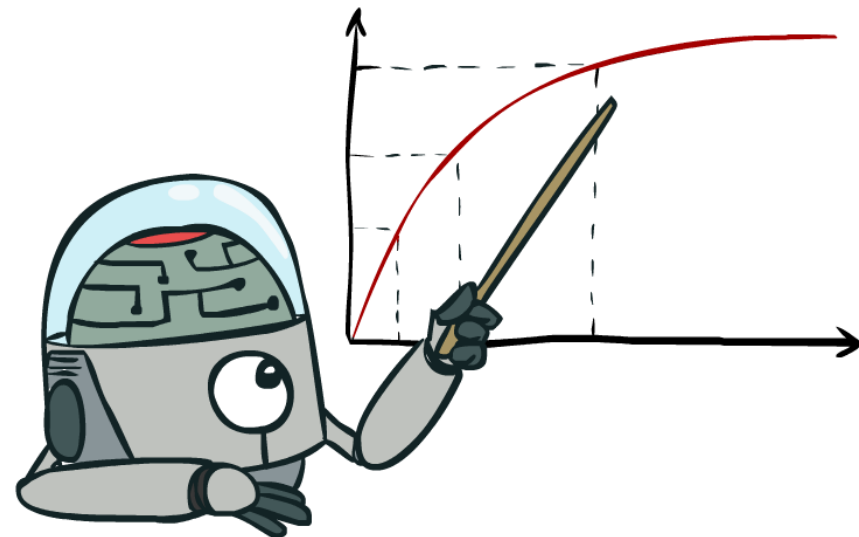
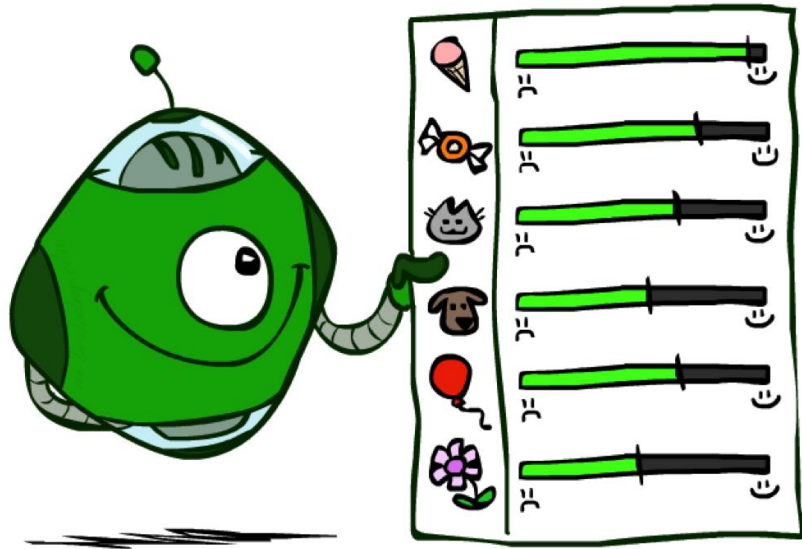
Acting Rationally

- Rational agent: Design AI systems that take optimal actions to achieve specific goals efficiently.
 - Focuses on goal-oriented behavior, not human-like thinking;
 - Chooses actions that maximize success, utility, or efficiency;
 - Operates based on outcomes, not emotions or intuition.



Acting Rationally

- A **rational agent** takes actions that optimally achieve its goals.
 - These goals depend on the specific application and are defined by the **utility** of possible outcomes.
 - To act rationally means to **maximize** expected utility—choosing actions that lead to the most beneficial results.



Acting Rationally

- **Rationality** is the ability of an agent to make decisions that maximize expected performance based on available information and defined goals.
 - It focuses solely on the decisions or actions taken, not the cognitive processes behind them.
 - Understanding the origins of cognition is not necessary for rational behavior.



Perfect Rationality

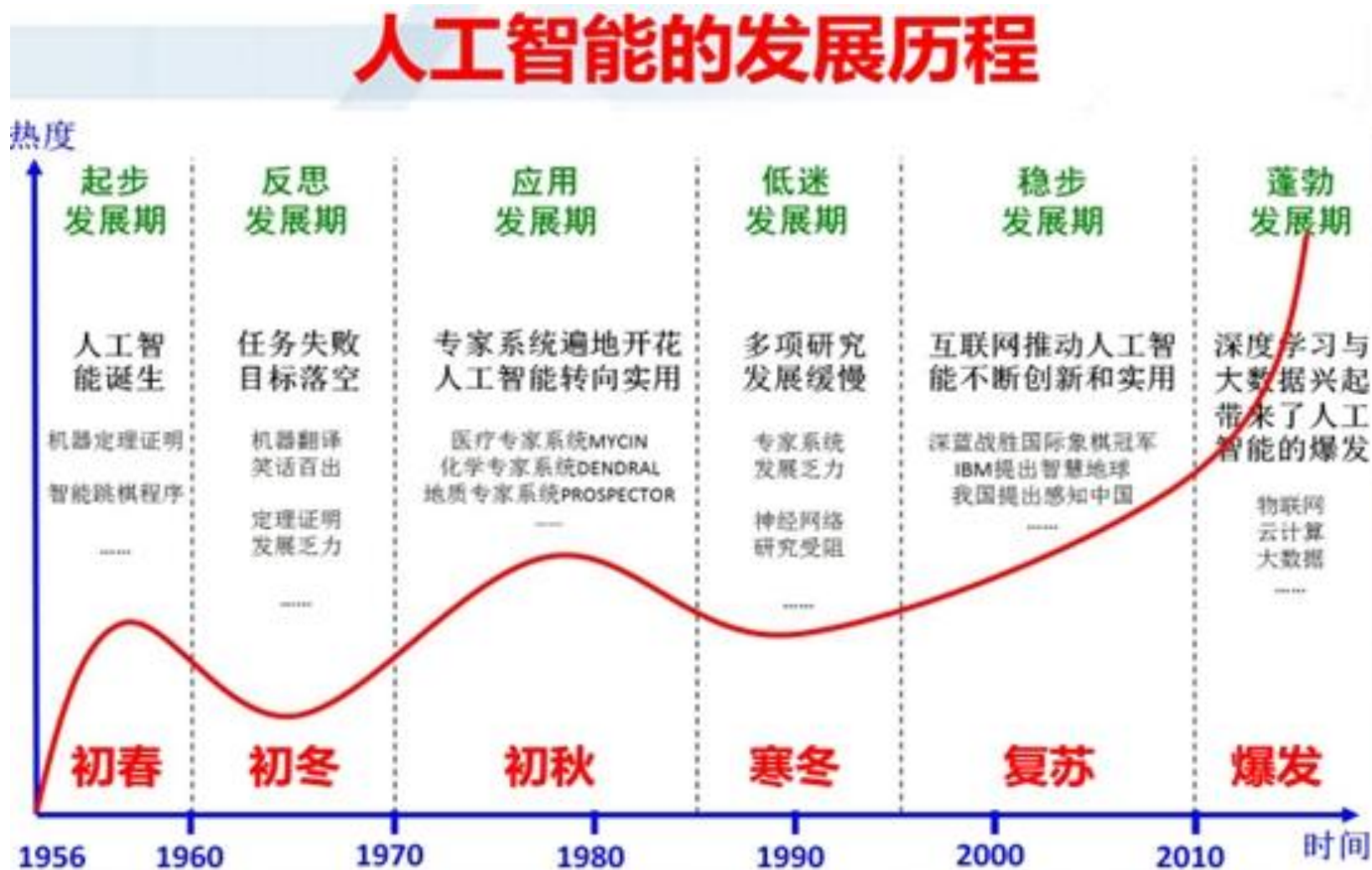
Assumes complete knowledge and unlimited computational resources to always choose the optimal action.

Bounded Rationality

Recognizes real-world limits—agents make good-enough decisions under constraints like time, data, or processing power.

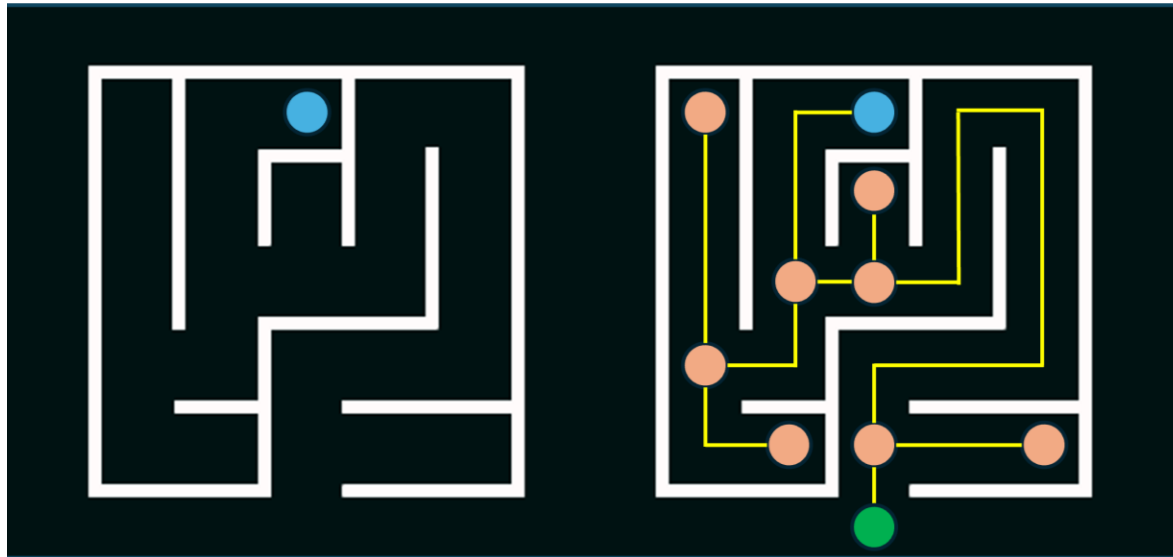


A Brief History of AI



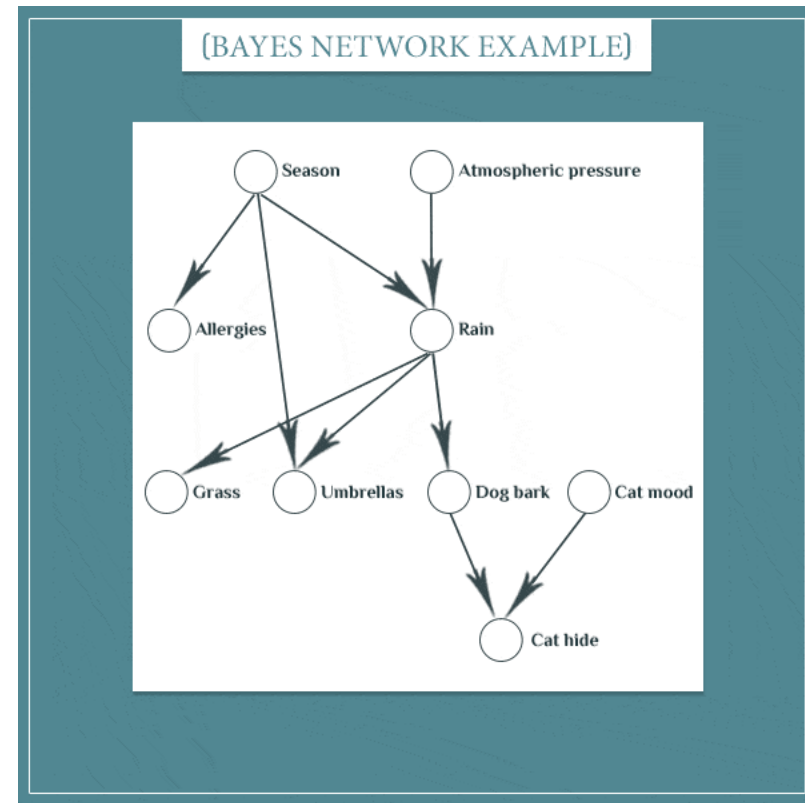
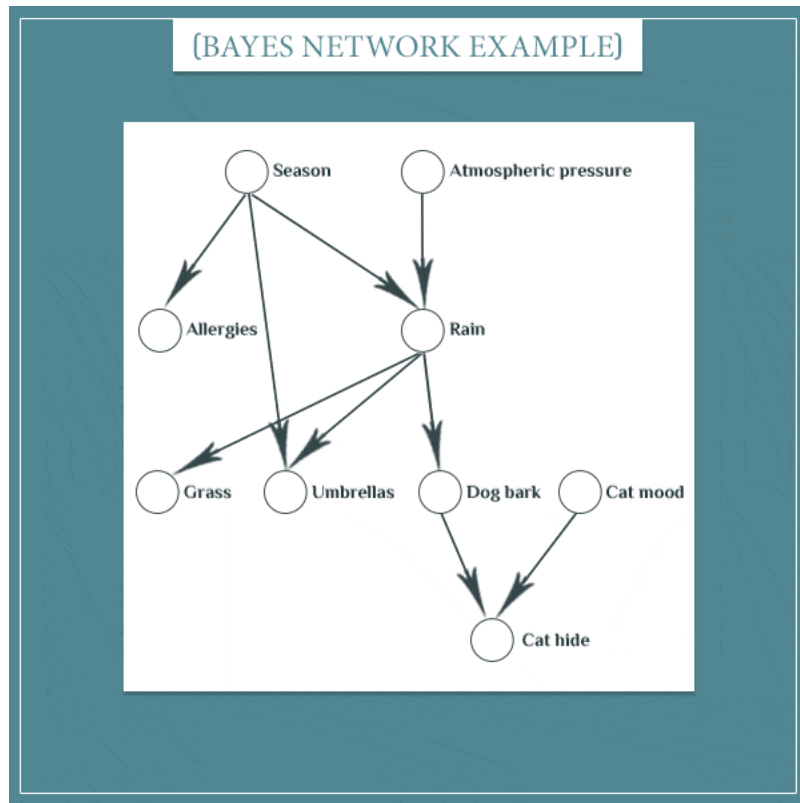
Search

- AI systems explore state spaces to find optimal solutions through algorithms like A*, depth-first, and breadth-first search.



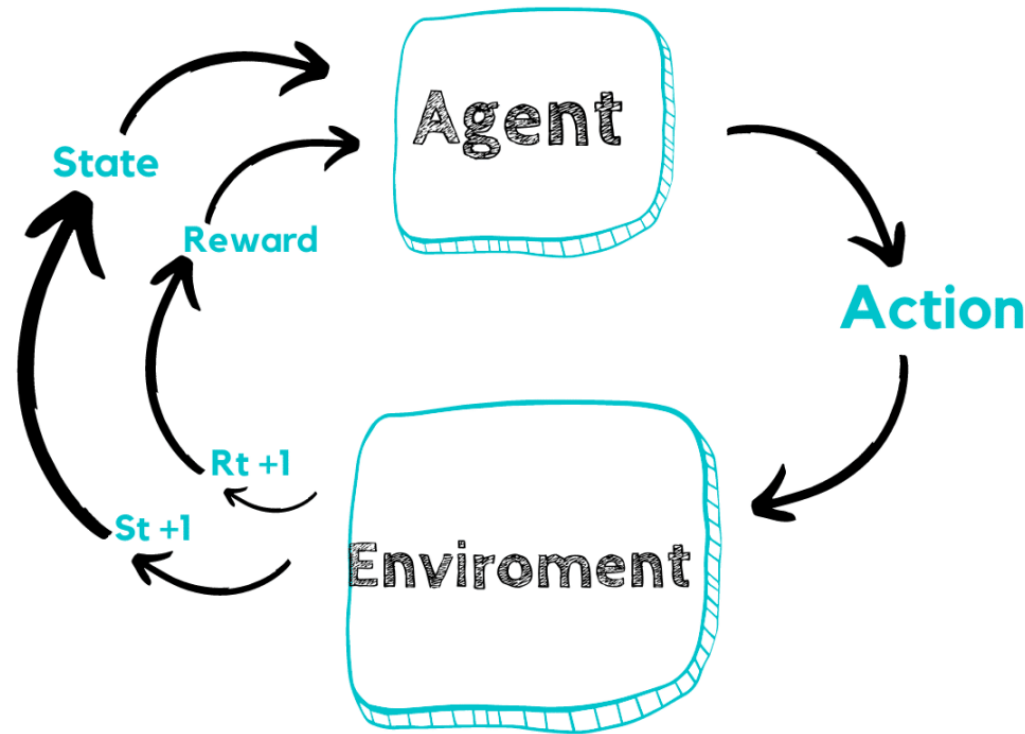
Bayesian Theory

- Bayesian methods help AI reason under uncertainty using probabilistic models such as Bayesian networks and Markov random fields.



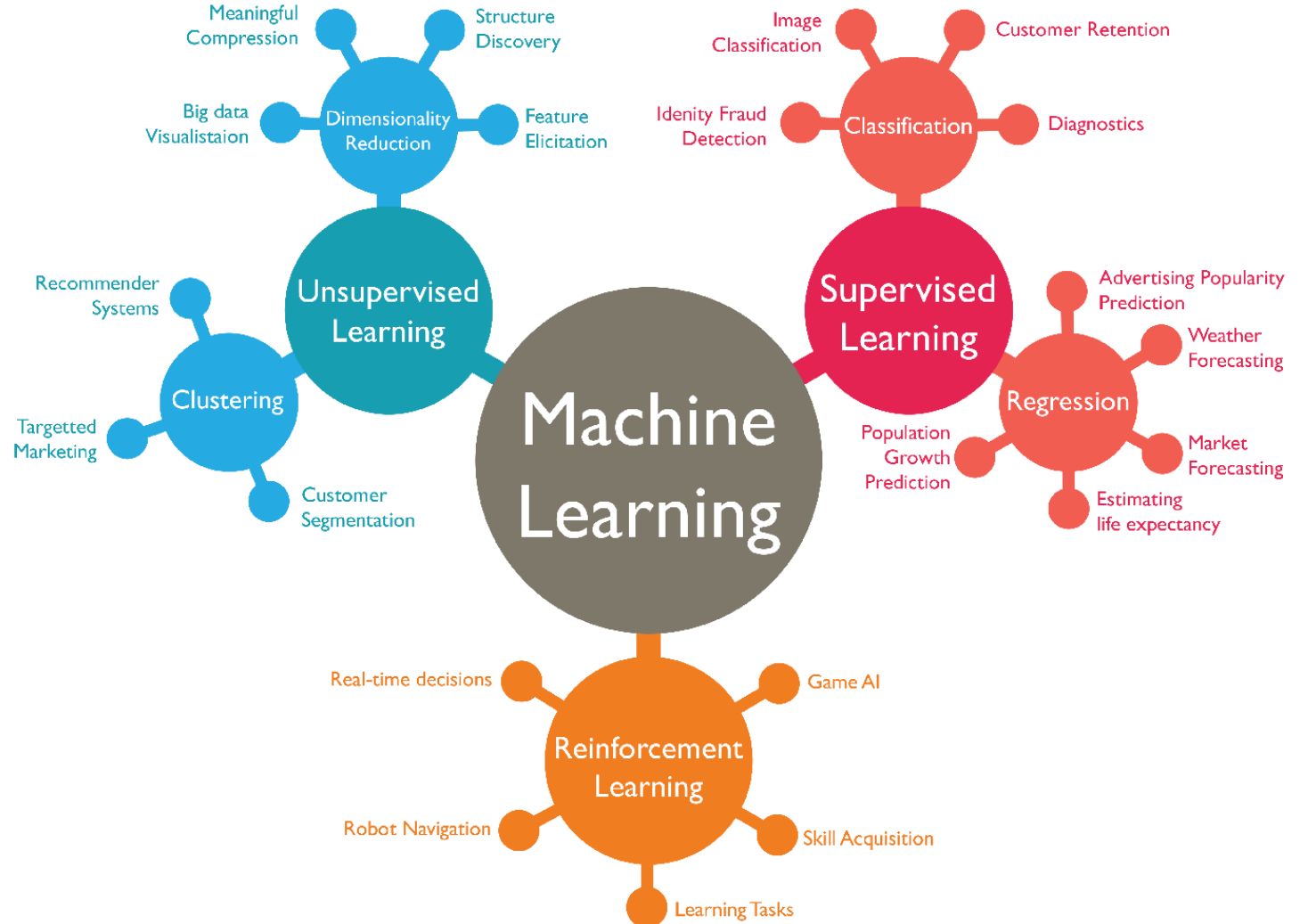
Reinforcement Learning

- Agents learn to make decisions by interacting with their environment and maximizing cumulative rewards over time.



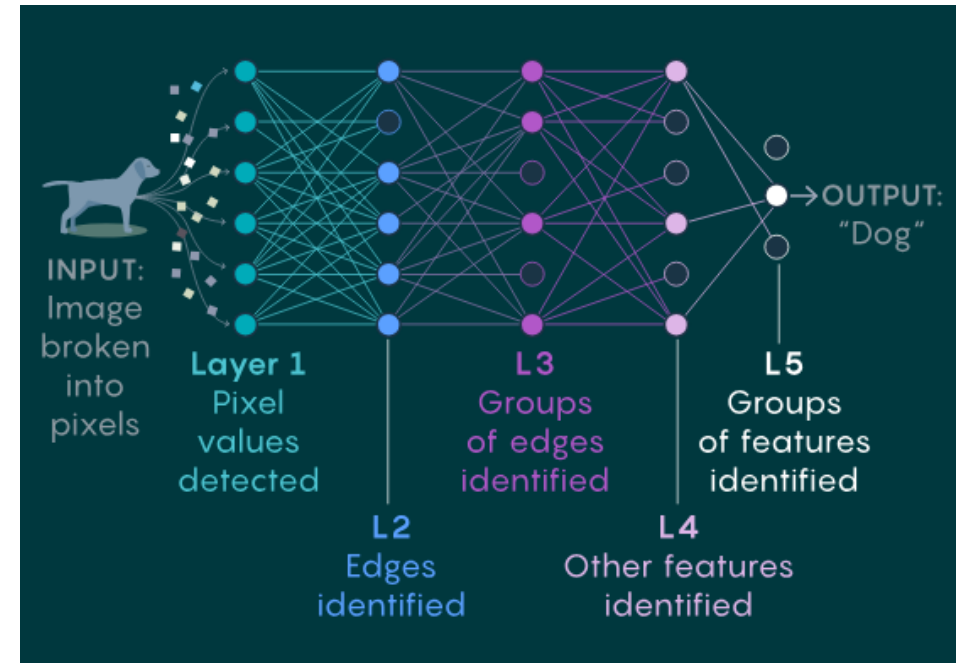
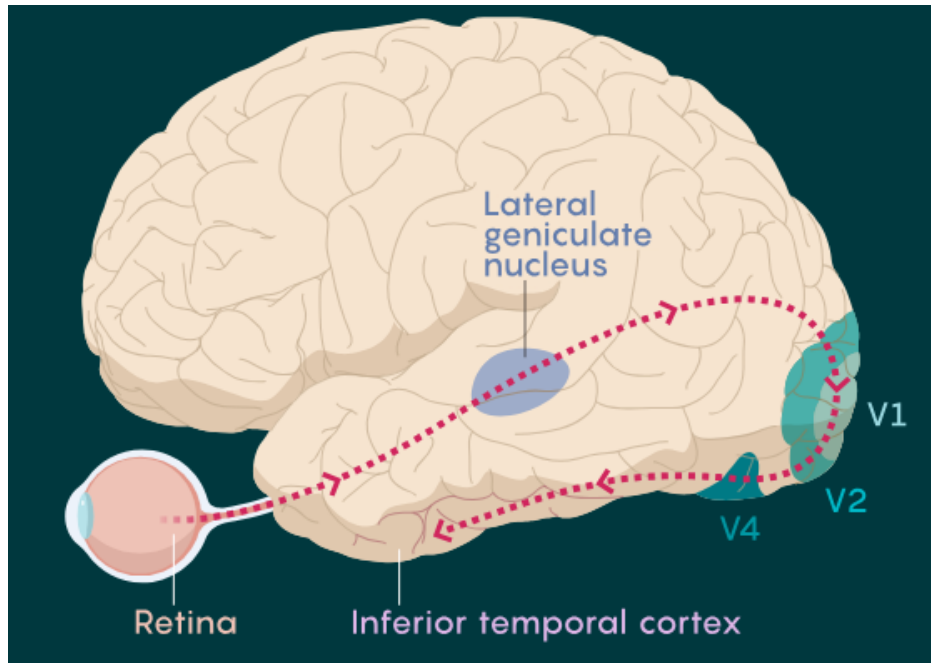
Machine Learning

- Algorithms learn patterns from data to perform tasks like classification, regression, and clustering without explicit programming.



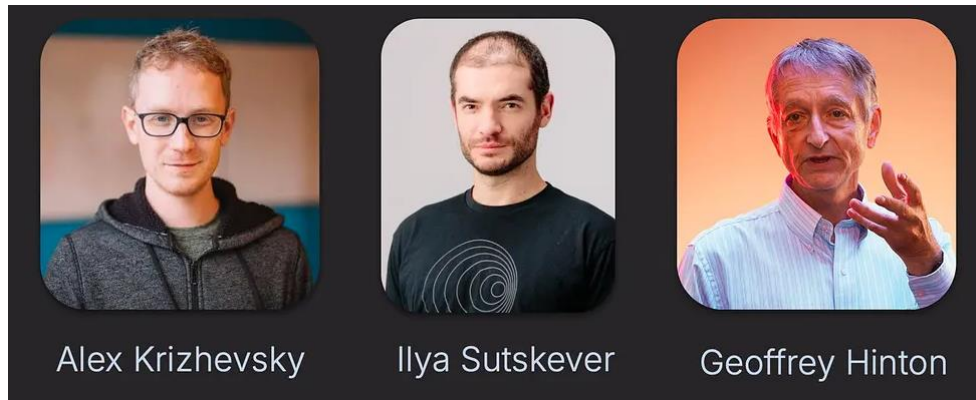
Deep Learning

- Neural networks with multiple layers enable AI to process complex data such as images, text, and speech with high accuracy.



Convolutional Neural Network

- In 2012, AlexNet, a deep convolutional neural network (CNN) developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) by a wide margin.



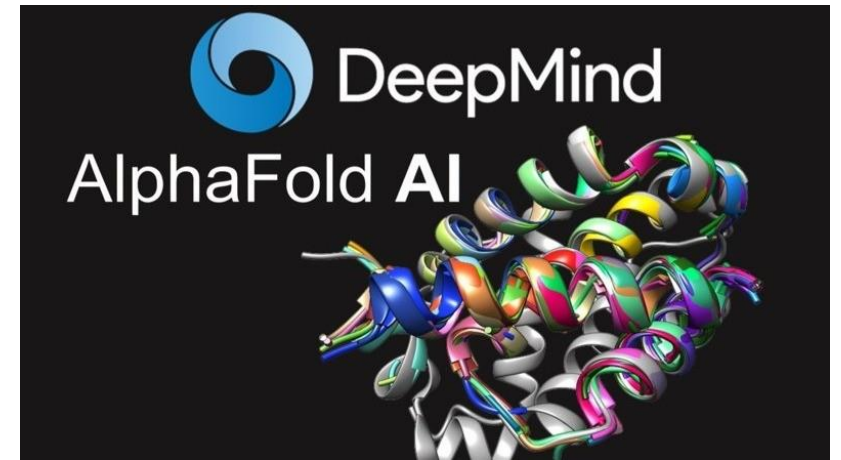
AlphaGo

- An AI system developed by DeepMind to master the game of Go.
 - 2016: Beat world champion Lee Sedol in a 5-game match (4 wins to 1);
 - Demonstrated superhuman performance and novel strategies beyond human intuition.



AlphaFold

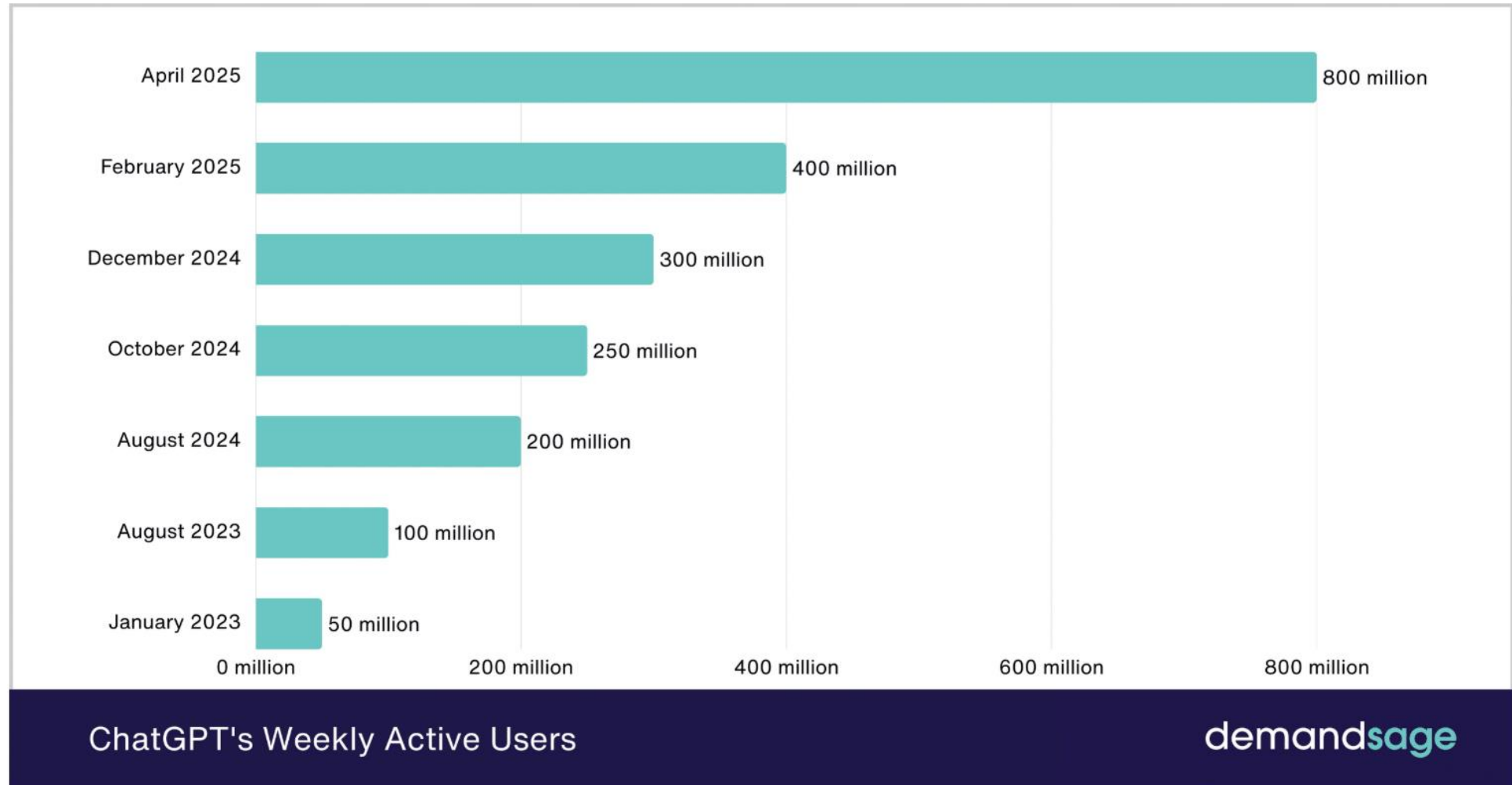
- An AI system developed by DeepMind that predicts the 3D structure of proteins from their amino acid sequences.
 - Proteins fold into complex shapes that determine their function;
 - Understanding these shapes is critical for drug discovery, disease research, and synthetic biology;
 - AlphaFold achieved atomic-level accuracy, rivaling experimental methods like X-ray crystallography.



"GPT Moment"

- GPT (Generative Pre-trained Transformer) is a large-scale neural network model based on the Transformer architecture.
- The release of GPT-3 marked a turning point in AI development:
 - Scale: 175 billion parameters;
 - Versatility: Capable of few-shot learning across diverse tasks;
 - Generalization: One model, many applications;
 - Accessibility: ChatGPT made advanced AI widely usable and visible to the public.

"GPT Moment"



Large Language Model (LLM)

- Recent advances in LLMs—especially in scale, reasoning, and multimodal capabilities—have accelerated at an extraordinary pace, reshaping both research and real-world applications.
 - GPT is a specific type of LLM.



How LLMs Empower You

- Act as a Personal Tutor
- Generate Code and Debug
- Summarize and Review Papers
- Brainstorm Project Ideas
- Simulate Peer Review
- Assist with Writing and Presentation
- Explore Frontiers

LLMs are collaborators in your learning journey.

Assignment - Zero

- Purpose: Learn AI with AI
- Set up a personal workflow where you can use AI tools (e.g., ChatGPT, DeepSeek) to:
 - Ask questions and clarify concepts;
 - Generate and debug code;
 - Draft and refine writing (e.g., reports, abstracts, documentation).

Assignment - Zero

- Submit a short report (1–2 pages) describing:
 - Which AI tools you chose and why;
 - How you configured or accessed them;
 - What tasks you tested (e.g., coding, Q&A, writing);
 - What worked well, what didn't, and what surprised you;
 - How you plan to integrate AI into your future learning.

Summary

- The trend of “AI + X” is transforming industries and daily life.
- Tools like LLM are reshaping how we learn, create, and interact.
- As students, we are entering an era where AI becomes a true collaborator.

Thank You!